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Research Summary

Data-Efficient and Human-Correctable Vision-Language-Action Systems for Real-Robot Manipulation

Research Focus

This proposal studies how language-conditioned manipulation policies can improve under a fixed budget of real-robot demonstrations and human operation time. The central premise is that informative correction after a failure can be more valuable than collecting another complete demonstration.

Established Preparation

My completed work in freeform optical automatic alignment combines optical modeling, structured camera acquisition, image-quality measurement, multi-objective optimization, six-degree-of-freedom physical adjustment, and repeated validation. In 10 repeated USTP alignment experiments, the final mean MTF exceeded 70%, with each run taking about 10 minutes.

Real-Robot Preliminary Study

I have established a real-robot preliminary platform based on an SO-101 leader-follower arm with fixed external and wrist RGB cameras. I collected 60 dual-view teleoperated episodes for three language-conditioned tabletop sorting instructions and constructed episode-level training, validation, and held-out test splits (48/6/6). I established an ACT visual-imitation baseline with 8/10 success on a fixed pick-and-place task, and fine-tuned a SmolVLA policy for the three-instruction setting.

The initial language-conditioned rollout produces target-directed motion but does not yet reliably complete the full grasp-and-place sequence under the limited demonstration budget. I treat this preliminary negative result as a research instrument rather than a purely engineering defect: it motivates controlled analysis of visual observability, grasp-stage action completion, initial-state variation, and the value of targeted corrective data.

PhD Research Direction

1. Under a matched human-time budget, can short corrective leader-teleoperation windows improve a dual-view language-conditioned policy more efficiently than collecting full demonstrations?
2. How do external and wrist-view observations affect failure modes in object selection, approach, grasp, lift, and placement?
3. How can simple observability and action constraints prevent unrecoverable action while preserving useful recovery opportunities?

Coherent Research Trajectory

Both optical and robotics settings require the same methodological discipline: define what an image measures, connect the measurement to a bounded physical action, record when the system is uncertain, and verify the outcome with repeated trials. The intended contribution is not an unrestricted language model for precision control; it is a testable hierarchy in which learned task-level action cooperates with visual measurement, constraints, and targeted human correction.

Project links: terra11113.github.io | huggingface.co/Terra11113